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HYPERURICEMIA: CURRENT INSIGHTS INTO PATHOGENESIS, ASSOCIATION WITH CHRONIC KIDNEY DISEASE, OXIDATIVE STRESS, AND PERSPECTIVES OF ANTIOXIDANT THERAPY

Abstract. *Hyperuricemia (HUr) is not only a consequence of impaired purine metabolism but also an independent risk factor for the development and progression of chronic kidney disease (CKD). This article provides a systematic review of the current pathophysiological mechanisms of renal tissue damage in HUr, particularly the activation of oxidative stress, inflammatory cascades, endothelial dysfunction, and fibrosis development. Special attention is given to uromodulin (Umod) as a biomarker of the tubular system and a potential mediator of nephropathological processes. The article also reviews current literature on the therapeutic potential of antioxidants, in particular glutathione and ubiquinone, in the context of pathogenetically targeted therapy for CKD associated with hyperuricemia. The review is based on sources published between 2021 and 2025 and does not include primary results from the authors' own clinical observations.* ***Conclusions****:* In the context of CKD, hyperuricemia requires a multifactorial diagnostic and therapeutic approa*ch, taking into account the role of oxidative stress and the prospects of antioxidant therapy.*

Keywords*: hyperuricemia; chronic kidney disease; uromodulin; oxidative stress; glutathione; ubiquinone; antioxidant therapy.*

INTRODUCTION

Hyperuricemia (HUr), defined as an elevation of serum uric acid levels above 6.8 mg/dL, is a common metabolic disorder traditionally associated primarily with the development of gout [1]. However, recent decades have demonstrated its much broader pathophysiological significance, particularly its role in the progression of chronic kidney disease (CKD), induction of oxidative stress, endothelial dysfunction, and impairment of target cell function [2,3]. HUr is not only a consequence of purine metabolism disorders but also an active factor capable of modifying the course of renal pathology and contributing to cardio-renal dysfunction [4,5].

The aim of this review article is to summarize current scientific evidence regarding the role of HUr in CKD pathogenesis, its association with oxidative stress and kidney tissue damage, and

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to substantiate the potential application of antioxidant therapy (particularly glutathione and ubiquinone) in the comprehensive management of such patients. Special attention is given to the role of uromodulin (Umod) as a potential marker of nephron functional status in this context.

Biochemistry and Pathophysiology of Hyperuricemia

Uric acid is the final product of purine catabolism in humans. Its metabolism, particularly the absence of uricase activity in the human body, is described in detail in the work of George et al. [1]. Due to the lack of the enzyme uricase, which converts uric acid to allantoin, its plasma concentration remains relatively high [6]. This leads to the accumulation of urates, especially in the presence of metabolic and excretory disorders [2, 3]. The main site of uric acid synthesis is the liver, where hypoxanthine and xanthine are oxidized to urates under the influence of xanthine oxidase [7]. The primary routes of uric acid excretion are renal (approximately 70%) and intestinal (around 30%).

The serum uric acid level is determined by the balance between its production and excretion. HUr may result either from excessive synthesis (due to consumption of purine-rich foods, alcohol, fructose) or from reduced excretion (due to decreased glomerular filtration rate or impaired function of transporters URAT1, GLUT9, ABCG2, whose role in the development of HUr has been detailed in the studies by Han et al. and Yang et al. [2, 3, 8, 9]).

Uric acid has a biphasic biological effect: in physiological concentrations, it acts as an antioxidant in plasma, but when elevated, it acquires prooxidant properties. Excessive urate levels activate NADPH oxidase, promote the formation of superoxide anions, and stimulate pro-inflammatory signaling pathways, particularly through activation of the NLRP3 inflammasome. These processes have been confirmed in experimental models demonstrating the active involvement of NADPH oxidase and inflammasome components in the pathogenesis of HUr [6, 8, 10, 11].

The result of these processes is the development of chronic inflammation, endothelial dysfunction, and damage to the renal vasculature. Decreased bioavailability of nitric oxide (NO) and activation of vascular inflammation have been described in the studies by Kuwabara et al. and Shubietah et al. [5, 9]. Urates also stimulate the proliferation of vascular smooth muscle cells, reduce the elasticity of the vascular wall, and decrease the availability of nitric oxide, which are additional factors contributing to the development of arterial hypertension and impairment of renal blood flow autoregulation [12, 13].

Hyperuricemia as a Risk Factor for Chronic Kidney Disease

Numerous epidemiological and clinical studies have confirmed the association between elevated uric acid levels and decreased glomerular filtration rate (GFR) [14, 15]. HUr is linked to the development of arterial hypertension, activation of the renin-angiotensin-aldosterone system, and direct damage to renal parenchyma [16, 17]. *See Fig. 1.*

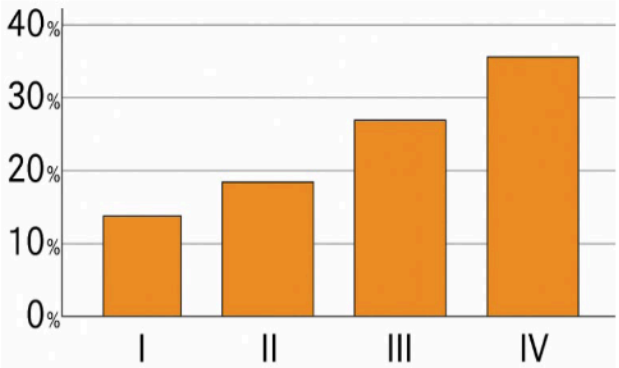

Fig. 1. Prevalence of HUr in different stages of CKD

One of the key pathogenetic mechanisms is the induction of oxidative stress, which leads to endothelial damage in renal capillaries, disruption of autoregulatory mechanisms of renal perfusion, ischemic changes, and progressive fibrosis [18]. Urates promote the production of proinflammatory cytokines such as IL-1β and TNF-α, activate the NLRP3 inflammasome, and initiate a cascade of interstitial inflammation [19, 20]. *See Fig. 2.*

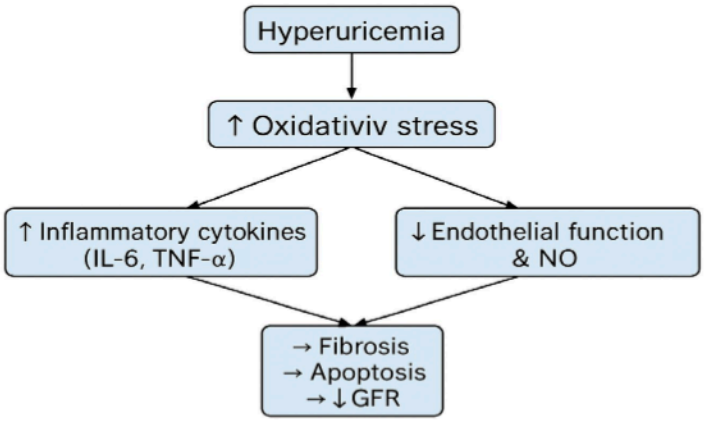

Fig. 2. Mechanisms of kidney injury in hyperuricemia

Studies in animal models and in patients with CKD have demonstrated that normalization of uric acid levels is associated with improved renal function and reduced progression of the disease [21, 22]. Moreover, HUr exerts an additive effect when combined with other metabolic disorders, particularly diabetes mellitus and obesity [23, 24].

Thus, HUr functions not only as a marker but also as an active participant in CKD pathogenesis, justifying the importance of early detection and correction of this condition in patients at risk of renal insufficiency. A general graphical representation of HUr-induced mechanisms of renal injury is provided in *Fig. 3.*

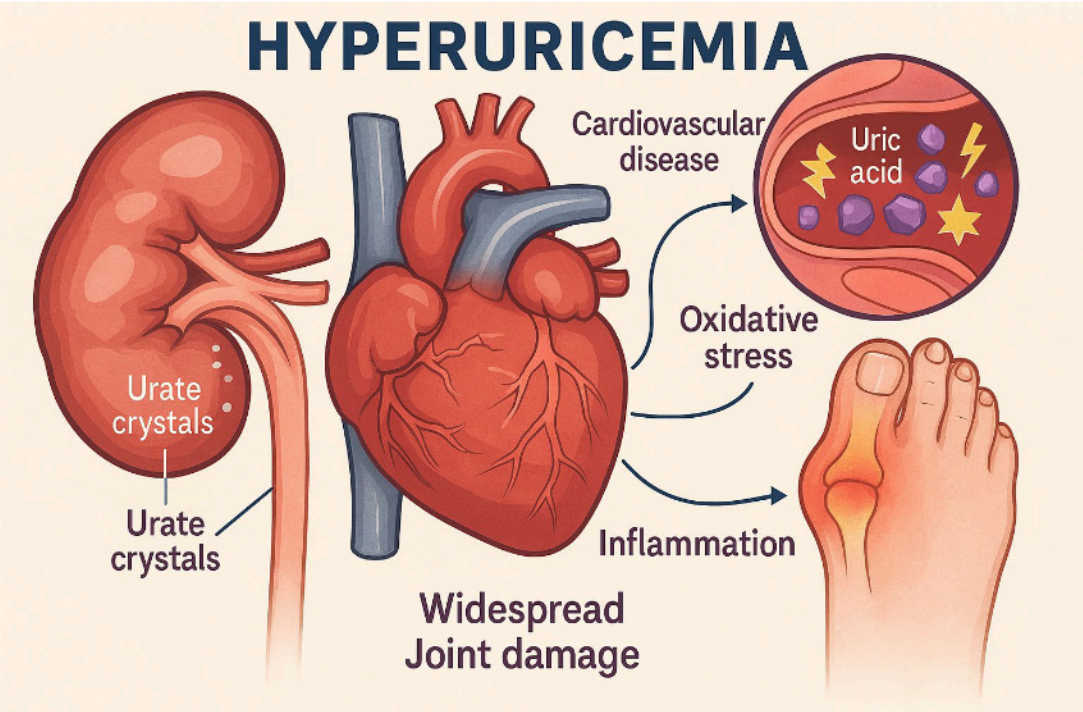

Fig. 3. Graphical illustration of mechanisms of kidney damage in Hur

The Role of Umod in CKD Pathogenesis

Uromodulin (Umod), also known as Tamm-Horsfall protein, is the most abundant protein in the urine of healthy individuals, synthesized in the thick ascending limb of the loop of Henle. It plays a key role in regulating water and electrolyte balance, providing antimicrobial protection of the urinary tract, and maintaining homeostasis in the renal tubules [25].

Under physiological conditions, Umod exerts antioxidant and anti-inflammatory effects, prevents the adhesion of urate crystals to the tubular epithelium, and modulates the immune response [26, 27]. However, in chronic kidney disease (CKD), especially under hyperuricemic conditions, its expression and secretion are decreased, leading to the loss of protective functions and increased susceptibility of nephrons to injury [28, 29].

A low urinary level of Umod is associated with deterioration of renal function and serves as an independent prognostic marker of CKD progression [30]. Moreover, mutations in the UMOD gene encoding Umod are associated with the development of autosomal dominant tubulointerstitial kidney disease, accompanied by hyperuricemia and early-onset gout [31, 32].

Therefore, Umod is considered a promising biomarker of tubular integrity and a potential target for future research in the context of CKD diagnosis and prognosis, particularly in patients with hyperuricemia. *See Fig. 4.*

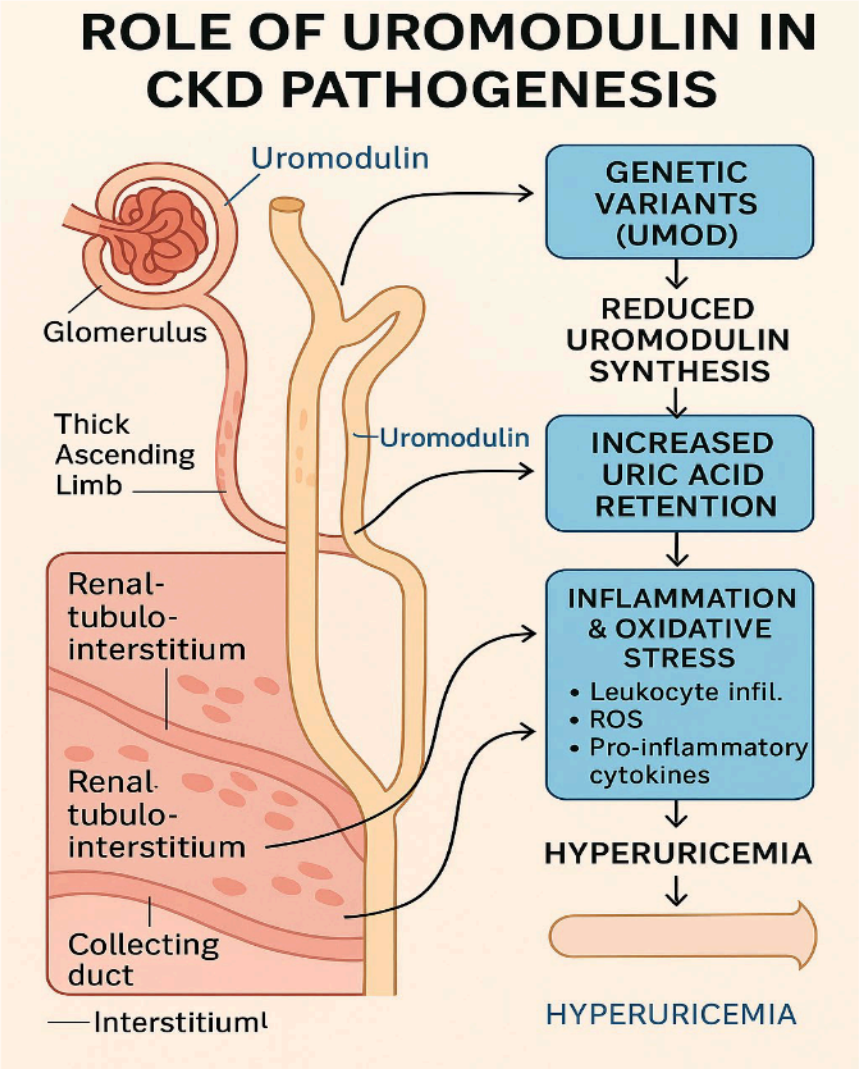

Fig. 4. The role of Umod in the pathogenesis of CKD

Antioxidant Therapy in HUr and CKD

Since oxidative stress is one of the key mechanisms of renal tissue damage in hyperuricemia (HUr), the use of antioxidants is considered a promising therapeutic strategy for chronic kidney disease (CKD). Reactive oxygen species (ROS), which are generated under conditions of elevated uric acid levels, cause structural cellular damage, mitochondrial dysfunction, apoptosis, and fibrosis [33, 34].

Glutathione (GSH) is a central intracellular antioxidant that participates in the neutralization of peroxides, free radicals, and detoxification reactions. In patients with CKD, a decrease in glutathione levels is observed, accompanied by elevated markers of oxidative stress, such as malondialdehyde (MDA), 8-hydroxy-2'-deoxyguanosine (8-OHdG), and protein carbonyls [35, 36]. Exogenous administration of glutathione or its precursors (e.g., N-acetylcysteine) has demonstrated beneficial effects in improving antioxidant status and stabilizing renal function in clinical studies [37].

Ubiquinone (coenzyme Q10, CoQ10) is another potent antioxidant that functions as an electron carrier in mitochondria and a participant in free radical scavenging reactions. Its deficiency is associated with mitochondrial dysfunction, activation of apoptosis, and inflammatory cascades in renal tissue [38]. Supplementation with CoQ10 helps to reduce oxidative stress, improve nephron energy metabolism, and downregulate the expression of pro-inflammatory genes [39].

Combined use of glutathione and coenzyme Q10 may produce a synergistic effect, as it targets several critical links in the pathogenesis of HUr and CKD: urate reduction, improvement of mitochondrial function, and attenuation of fibrosis and inflammation. *See Table 1.* In this context, the study of redox balance biomarkers, including uromodulin (Umod), appears promising as a potential indicator of antioxidant therapy effectiveness.

Table 1. Comparative characteristics of antioxidants

No	Antioxidant	Mechanism of Action	Potential Role in CKD
1	Glutathione	Neutralizes free radicals, restores thiol groups in proteins	Reduces oxidative tubular injury
2	Ubiquinone	Protects mitochondria, inhibits lipid peroxidation	Improves nephron energy metabolism
3	Vitamin C	Regenerates vitamin E, neutralizes superoxide anions	Maintains general antioxidant status

CONCLUSIONS

HUr is not only a marker of disordered purine metabolism but also a significant pathogenic factor in the development and progression of CKD. Its effects are mediated through oxidative stress activation, promotion of inflammation, ischemic injury to nephrons, and interstitial fibrosis. Umod dysfunction in the context of HUr further increases the vulnerability of renal tissue by impairing anti-inflammatory defense and electrolyte regulation.

Modern approaches to the treatment of hyperuricemia (HUr) should include not only urate-lowering agents such as allopurinol or febuxostat, but also agents with combined actions, particularly SGLT2 inhibitors, which reduce serum uric acid levels and exert nephroprotective effects [40], as well as pathogenetically justified antioxidant support. Glutathione and ubiquinone show promise in reducing oxidative stress, improving mitochondrial function, and stabilizing CKD progression, particularly in patients with elevated uric acid levels.

Therefore, combining antioxidant and urate-lowering therapy may represent an effective strategy for slowing CKD progression. Further studies of biomarkers, including Umod, are warranted as

indicators of treatment efficacy and renal dysfunction progression. Large-scale clinical trials are needed to justify the long-term effectiveness and safety of such therapeutic approaches.

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Гіперурикемія: сучасні уявлення про патогенез, взаємозв'язок з хронічною хворобою нирок, оксидативним стресом і перспективи антиоксидантної терапії

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Анотація. *Гіперурикемія (HUr) є не лише наслідком порушеного пуринового метаболізму, а й незалежним чинником ризику розвитку та прогресування хронічної хвороби нирок (ХХН). У статті здійснено систематизований огляд сучасних патофізіологічних механізмів ураження ниркової тканини при HUr, зокрема активації оксидативного стресу, запальних каскадів, ендотеліальної дисфункції та розвитку фіброзу. Особливу увагу приділено уромодуліну (Umod) як біомаркеру стану канальцевого апарату і потенційному медіатору нефропатологічних процесів. Окремо розглянуто сучасні літературні дані щодо можливостей використання антиоксидантів, зокрема глутатіону та убіхінону, в контексті патогенетично орієнтованої терапії ХХН на тлі гіперурикемії. Огляд базується на джерелах 2021–2025 рр. і не містить первинних результатів власних клінічних спостережень.* ***Висновки:*** *HUr в контексті ХХН потребує мультифакторного діагностично-терапевтичного підходу з урахуванням ролі оксидативного стресу та перспектив антиоксидантної терапії.*

Ключові слова: *гіперурикемія; хронічна хвороба нирок; уромодулін; оксидативний стрес; глутатіон; убіхінон; антиоксидантна терапія.*

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Author Contributions:
L. D. Denova *– was responsible for data collection and analysis, manuscript writing, search and review of relevant scientific literature, and preparation of the manuscript for publication;*
S. V. Kushnirenko *– conceptualization, methodology.*

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